Temporary Capture in the Earth-Moon Circular Restricted Three-Body Problem (CR3BP)

ASEN 6062: Celestial Mechanics

Nicole James

April 24, 2026

Ann and H.J. Smead
Aerospace Engineering Sciences
UNIVERSITY OF COLORADO BOULDER

1 Abstract

This work examines temporary ballistic capture in the Earth–Moon Circular Restricted Three-Body Problem using a Hamiltonian framework in the normalized synodic frame. Linear stability analysis about the L_1/L_2 equilibrium points and associated State Transition Matrices reveal how Lyapunov orbits and their invariant manifolds govern capture, residence time, and escape.

2 Nomenclature

$A(t)$	Jacobian matrix
H	Hamiltonian
i	Imaginary unit $\sqrt{-1}$
J	Canonical symplectic matrix
L	Lagrangian
n	Mean motion
r	Distance $\sqrt{x^2 + y^2 + z^2}$
T	Kinetic energy
U	Potential energy, effective potential
U_{ij}	Second partial derivatives of the effective potential
W^s, W^u	Stable and unstable invariant manifolds
$\mathbf{x}$	State vector, $[\vec{x}, \dot{\vec{x}}]^T$
α	Real unstable eigenvalue at L_1/L_2
β	Center frequency at L_1/L_2
λ	Generic eigenvalue
θ	Angular position
μ	Constant mass parameter $\frac{m_2}{m_1+m_2}$
ω	Angular velocity
$\Phi(t, t_0)$	State transition matrix (STM)

Contents

1 Abstract **2**

2 Nomenclature **2**

3 Introduction **4**

4 Circular Restricted Three-Body Problem Dynamics **4**
- 4.1 Setup and Assumptions 4
- 4.2 Massless Particle Dynamics 4
  - 4.2.1 Motion in Inertial Frame 4
  - 4.2.2 Lagrangian Derivation 5
  - 4.2.3 Hamilton Derivation 6
- 4.3 Jacobi Integral and Zero Velocity Curves 6
- 4.4 State Transition Matrix 8
  - 4.4.1 STM Derivation 8
  - 4.4.2 The L_1 and L_2 Equilibrium Points 10
  - 4.4.3 Lyapunov Orbits 10

5 Simulation and Results **10**
- 5.1 Determination of Initial Conditions 10
- 5.2 Calculations and Verification of Numerical Results 12
- 5.3 Presentation of Results 12

6 Discussion and Future Investigation **13**

7 Conclusion **14**

8 References **14**

List of Figures

1 Zero Velocity Curves for Various Jacobi Constants in the Earth-Moon System (Full View) 7
2 Zero Velocity Curves for Various Jacobi Constants in the Earth-Moon System (Moon Hill Region Zoom) 8
3 Lyapunov Orbits near L_1 and L_2 10
4 Lyapunov Manifolds near L_1 and L_2 red=W^s, blue=W^u 11
5 Forbidden Regions for Three Jacobi Constants 11
6 Simulated Trajectories for Third Body with Three Jacobi Constants Panels correspond to C=C(L_1), C=3.1804, C=C(L_2) 12
7 Sampling of Trajectories Originating Near L_1 red=captured, blue=entry,orange=escape 13

3 Introduction

This work investigates temporary satellite capture in the Earth–Moon Circular Restricted Three-Body Problem using a Hamiltonian dynamical systems framework. The dynamics of the system are described in Section 4. The equations of motion are derived in the inertial and normalized synodic frame in sections 4.1 and 4.2. They are analyzed through the Jacobi integral, zero-velocity curves, and linear stability about the collinear equilibrium points in sections 4.3. State Transition Matrices are used to characterize local dynamics near L_1/L_2 with the eigenvalues revealing the saddle–center structure that gives rise to Lyapunov orbits and their invariant manifolds in section 4.4. Temporary capture is defined as motion that enters the lunar region along stable manifolds, remains confined in the Moon's Hill sphere for finite time, and escapes along unstable manifolds. Numerical simulations and results are presented in section 5. The discussion of findings and potential future investigations are discussed in section 6.

4 Circular Restricted Three-Body Problem Dynamics

4.1 Setup and Assumptions

The solution to finding the trajectories of particles which result in a temporary lunar capture can be best modeled using the Circular Restricted Three Body Problem (CR3BP) in a uniformly rotating (synodic) frame[6]. The CR3BP models the planar motion of a massless test particle or third body under the gravitational influence of two primary bodies with masses m_1 and m_2 where $m_1 > m_2$. The primaries are assumed to move on circular orbits about their common barycenter, while the third body has negligible mass and therefore does not perturb the motion of the primaries. This idealization captures key dynamics of systems with a strong mass hierarchy (e.g., spacecraft motion in the Earth–Moon or Sun–Earth environments), while retaining the essential non-integrability of three-body gravitational motion; in general, trajectories of the test particle must be propagated numerically. Analysis is performed in a synodic reference frame whose origin is fixed at the system barycenter and whose angular velocity matches the mean motion of the primaries so that both primaries appear stationary and form a relative equilibrium. The distance between the primaries is set to unity, the sum of the primary masses is unity and the system is characterized by the constant mass parameter $\mu = \frac{m_2}{m_1+m_2}$. In this frame, the primaries lie on the x-axis at fixed locations (-μ, 0, 0) and (1-μ, 0, 0) while a point co-rotating with the frame has zero velocity. The resulting equations of motion include both Coriolis and centrifugal terms.

4.2 Massless Particle Dynamics

4.2.1 Motion in Inertial Frame

In an inertial frame defined with the barycenter of the primary bodies as the origin, the location of the infinitesimal third body is defined by[6]:

$$r_1^2 = (x_1 - x)^2 + (y1 - y)^2 + z^2 = (x + \mu)^2 + y^2 + z^2 \tag{1}$$

$$r_2^2 = (x_2 - x)^2 + (y_2 - y)^2 + z^2 = (x - 1 + \mu)^2 + y^2 + z^2 \tag{2}$$

where x, y and z are the coordinates of the third body, x_1, y_1 and z_1 are the coordinates of the primary, x_2, y_2 and z_2 are the coordinates of the secondary, r_1 is the relative distance between the first and third bodies, r_2 is the relative distance between the second and third bodies and μ is the mass parameter. The equations of motion for the third body in this inertial frame are as follows, where $\ddot{\boldsymbol{r}}$, $\ddot{x}$, $\ddot{y}$ and $\ddot{z}$ are the respective accelerations:

$$\ddot{\boldsymbol{r}} = -(1-\mu)\frac{\boldsymbol{r} - \boldsymbol{r_1(t)}}{r_1^3} - \mu\frac{\boldsymbol{r} - \boldsymbol{r_2(t)}}{r_2^3} \tag{3}$$

$$\ddot{x} = -(1-\mu)\frac{x - x_1(t)}{r_1^3} - \mu\frac{x - x_2(t)}{r_2^3} \tag{4}$$

$$\ddot{y} = -(1-\mu)\frac{y - y_1(t)}{r_1^3} - \mu\frac{y - y_2(t)}{r_2^3} \tag{5}$$

$$\ddot{z} = -(1-\mu)\frac{z}{r_1^3} - \mu\frac{z}{r_2^3} \tag{6}$$

4.2.2 Lagrangian Derivation

In an inertial frame, the gravitational potential depends explicitly on time because the primaries are moving. Consequently, the Lagrangian is time-dependent and the system does not have a conserved energy integral nor time-invariant equilibrium points in inertial coordinates[6]. The Lagrangian in the inertial frame, L_I, is:

$$L_I = f(\boldsymbol{r}, \dot{\boldsymbol{r}}, t) = T - U = \frac{1}{2}||\dot{r}^2|| + \frac{\mu_1}{||\boldsymbol{r} - \boldsymbol{r_1}(t)||} + \frac{\mu_2}{||\boldsymbol{r} - \boldsymbol{r_2}(t)||} \tag{7}$$

where T is the kinetic energy, U is the potential energy, $\boldsymbol{r}$ is the position of the test particle, $\boldsymbol{r_1}$ and $\boldsymbol{r_2}$ are the positions of the primaries varying over time. To obtain an autonomous formulation, the system is transformed into a synodic frame as defined in the previous section. In this frame, the angular velocity $\boldsymbol{\Omega} = n\hat{z}$ where n is the mean motion of the primaries and is equal to one. The system is parameterized by μ, primaries are fixed and the state of the test particle must be numerically computed over time. In this frame,

$$\boldsymbol{r_1} = (-\mu, 0, 0) \tag{8}$$

$$\boldsymbol{r_2} = (1-\mu, 0, 0) \tag{9}$$

$$\boldsymbol{q} = (x, y, z) \tag{10}$$

$$\dot{\boldsymbol{q}} = \dot{\boldsymbol{r_I}} + \boldsymbol{\Omega} \times \boldsymbol{r} \tag{11}$$

where $\boldsymbol{r_1}$ and $\boldsymbol{r_2}$ are fixed at their locations, $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$ are the position and velocity of the third body, respectively, and $\dot{\boldsymbol{r_I}}$ is the velocity of the third body in the inertial frame. With $||\boldsymbol{\Omega}|| = 1$ and $\dot{\Omega} = 0$ the Lagrangian then becomes time invariant:

$$L = f(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \frac{1}{2}||\dot{\boldsymbol{q}}||^2 + \boldsymbol{\Omega} \times (\boldsymbol{q} \times \dot{\boldsymbol{q}}) + \frac{1}{2}||\boldsymbol{\Omega} \times \boldsymbol{q}||^2 + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} \tag{12}$$

$$L = \frac{1}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + (x\dot{y} - \dot{x}y) + \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} \tag{13}$$

where $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, r_1, and r_2 are defined as above. The equations of motion are then derived from the time derivative of the Lagrangian:

$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{\boldsymbol{q}}}) - \frac{\partial L}{\partial \boldsymbol{q}} = 0 \tag{14}$$

and become:

$$\ddot{x} - 2\dot{y} = x - \frac{1-\mu}{r_1^3}(x + \mu) - \frac{\mu}{r_2^3}(x - 1 + \mu) \tag{15}$$

$$\ddot{y} + 2\dot{x} = y - \frac{1-\mu}{r_1^3}y - \frac{\mu}{r_2^3}y \tag{16}$$

$$\ddot{z} = -\frac{1-\mu}{r_1^3}z - \frac{\mu}{r_2^3}z \tag{17}$$

where (x, y, z) are the coordinates, $(\dot{x}, \dot{y}, \dot{z})$ and $(\ddot{x}, \ddot{y}, \ddot{z})$ are the respective velocity and acceleration of the third body at any point in time.

4.2.3 Hamilton Derivation

The Hamiltonian form defines the dynamical system's solution properties and reveals the structure of the CR3BP. It is central to analyzing conserved quantities and phase-space dynamics. The Hamiltonian is obtained via a Legendre transformation of the synodic-frame Lagrangian, equation 13. The canonical momenta, p, are associated with the coordinates, $\boldsymbol{q}$, defined as $\boldsymbol{p} = \frac{\partial L}{\partial \dot{\boldsymbol{q}}}$. The momenta are:

$$p_x = \dot{x} - y \tag{18}$$
$$p_y = \dot{y} + x \tag{19}$$
$$p_z = \dot{z} \tag{20}$$

with velocities expressed as:

$$\dot{x} = p_x + y, \dot{y} = p_y - x, \dot{z} = p_z \quad (\dot{\boldsymbol{q}}) \tag{21}$$

The Hamiltonian is defined by:

$$H(\boldsymbol{q}, \boldsymbol{p}) = \boldsymbol{p} \cdot \dot{\boldsymbol{q}} - L \tag{22}$$

Substituting the above relations yields[6]:

$$H = \frac{1}{2}(p_x^2 + p_y^2 + p_z^2) + p_x y - p_y x - \frac{1-\mu}{r_1} - \frac{\mu}{r_2} \tag{23}$$

where r_1 and r_2 are defined in equations 1 and 2. The Hamiltonian takes on the form:

$$\dot{\boldsymbol{q}} = \frac{\partial H}{\partial \boldsymbol{p}} = \boldsymbol{p} - \boldsymbol{\Omega} \times \boldsymbol{q} \tag{24}$$
$$\dot{\boldsymbol{p}} = -\frac{\partial H}{\partial \boldsymbol{q}} = -\boldsymbol{\Omega} \times \boldsymbol{p} + \frac{\partial U}{\partial \boldsymbol{q}} \tag{25}$$

Explicitly[6],

$$\dot{p}_x = \frac{\partial H}{\partial x} = p_y - [(1-\mu)\frac{x+\mu}{r_1^3} + \mu\frac{x-1+\mu}{r_2^3}] \tag{26}$$
$$\dot{p}_y = -\frac{\partial H}{\partial y} = -p_x - [(1-\mu)\frac{y}{r_1^3} + \mu\frac{y}{r_2^3}] \tag{27}$$
$$\dot{p}_z = \frac{\partial H}{\partial z} = -[(1-\mu)\frac{z}{r_1^3} + \frac{\mu z}{r_2^3}] \tag{28}$$

and $\dot{\boldsymbol{q}}$ is defined in equation 21. Differentiating the position equations, 1 and 2 and substituting the momenta equations restores the equations of motion defined in Section 4.2.2, confirming consistency between the Lagrangian and Hamiltonian formulations.

4.3 Jacobi Integral and Zero Velocity Curves

Because the Lagrangian is time-invariant in the synodic frame, equation 13, the Legendre transform yields a conserved integral of motion, called the Jacobi integral. The Hamiltonian, H, is constant

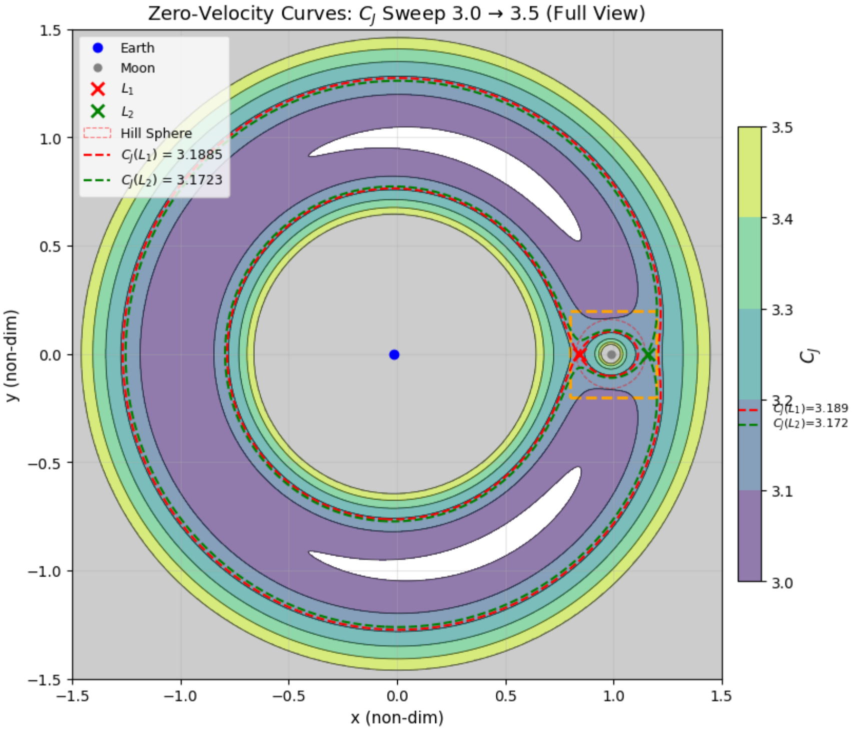

Figure 1: Zero Velocity Curves for Various Jacobi Constants in the Earth-Moon System (Full View)

along trajectories where $\frac{dH}{dt} = 0$. To relate this to the Jacobi integral, the effective potential is defined as:

$$U(x, y, z) = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} \tag{29}$$

Expressing the Hamiltonian in terms of velocity yields:

$$H = \frac{1}{2}v^2 - U(x, y, z) \tag{30}$$

Multiplying that by -2 defines the Jacobi integral:

$$C = 2U(x, y, z) - v^2 \tag{31}$$

which relates to the Hamiltonian by[6]:

$$C = -2H \tag{32}$$

Motion is restricted to regions where $2U(x, y, z) \geq C$. The boundary $2U(x, y, z) = C$ defines the zero-velocity surface (ZVS) which yields the zero-velocity curves (ZVCs) when z=0[1]. Shown in figures 1 and 2. These curves define accessible and forbidden regions for given values of the Jacobi constant. These regions constrain transport trajectories between the Earth and Moon and govern capture and escape through the L_1 and L2 gateways.

Now that the global energy constraints are known, a definition of temporary lunar capture is established. In planetary dynamics, a particle is said to be temporarily captured by the secondary body if it enters the Moon's Hill sphere with sufficiently low relative velocity such that it becomes gravitationally bound for a finite duration before eventual escape. The energy is given by: [1]

$$E_{rel} = \frac{1}{2}v_{rel}^2 - \frac{(1-\mu)}{r_{rel}} \tag{33}$$

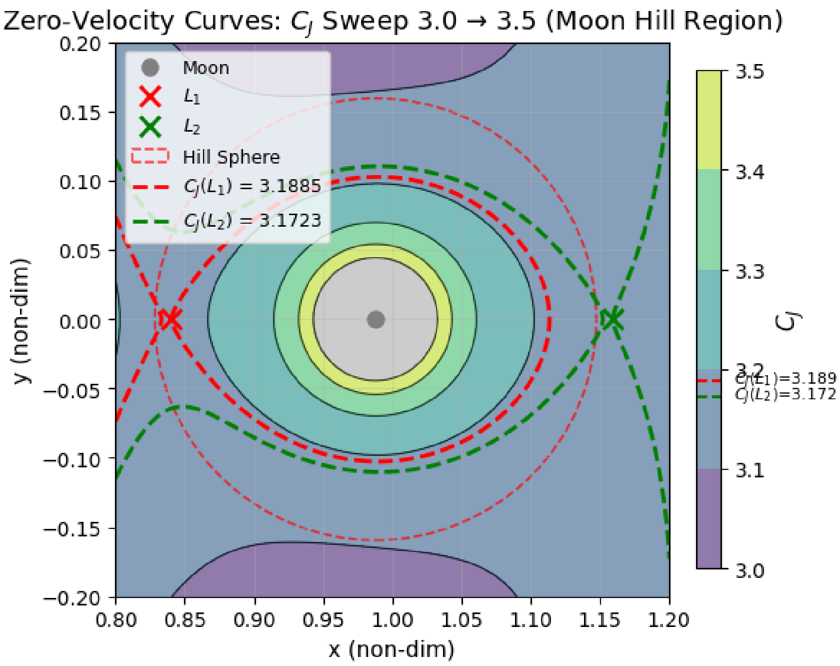

Figure 2: Zero Velocity Curves for Various Jacobi Constants in the Earth-Moon System (Moon Hill Region Zoom)

A trajectory is classified as temporarily captured when $E_{rel} < 0$ over a continuous time interval while the particle remains within the lunar Hill sphere.

4.4 State Transition Matrix

While the Jacobi integral and ZVCs provide global constraints on the massless third body, they do not characterize the local behavior of a trajectory at any particular point. ZVCs identify if motion is permitted or forbidden in a region, but they do not give any information on the actual trajectory or how neighboring trajectories evolve. Temporary lunar capture can occur in regions where the ZVCs allow access to the Moon's Hill sphere through the L_1 and L_2 gateways but do not guarantee long-term capture. This behavior is governed by the local stability properties of the system solutions. To quantify this local behavior, the state transition matrix (STM) of the linearized system along a nominal trajectory is evaluated. The STM identifies the stability of the system to perturbations and helps distinguish temporary capture from escape dynamics.

4.4.1 STM Derivation

At this point, the system will be restricted to the planar CR3BP to simplify the analysis of the STM which removes the z coordinates and becomes a 4x4 matrix. Starting with Lagrangian equations:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\boldsymbol{q}}}\right) - \frac{\partial L}{\partial \boldsymbol{q}} = 0 \tag{34}$$

with second order ordinary differential equation $\ddot{q} = a(q, \dot{q})$

$$\ddot{\boldsymbol{q}} = J\nabla H(\boldsymbol{q}, \boldsymbol{p}) \tag{35}$$

where J is the symplectic matrix:

$$\begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} \quad I = I_{2x2} \tag{36}$$

Linearizing the equations of motion about a nominal solution with small perturbations, $\delta\boldsymbol{x}(t)$ gives[1]:

$$\delta\dot{\boldsymbol{x}} = \frac{\delta}{\delta x}(J\nabla H(x))\Big|_{x(t)}\delta x \tag{37}$$

$$= J\nabla^2 H(x(t))\delta x \tag{38}$$

where $\nabla^2 H$ is the Hessian of the Hamiltonian:

$$\nabla^2 H = \begin{bmatrix} H_{qq} & H_{qp} \\ H_{pq} & H_{pp} \end{bmatrix} = \begin{bmatrix} H_{xx} & H_{xy} & 0 & -1 \\ H_{xy} & H_{yy} & 1 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \tag{39}$$

where $H_{pp} = I$, H_{qp} is the Coriolis coupling and H_{qq} contains the second derivative of the effective potential U_{xx}, U_{yy} and U_{xy}. The STM $\Phi(t)$ maps initial perturbations:

$$\delta\boldsymbol{x}(t) = \Phi(t, t_0)\delta\boldsymbol{x}(t_0) \tag{40}$$

and satisfies:

$$\dot{\Phi}(t) = A(t)\Phi(t), \quad \Phi(t_0) = I_{2x2} \tag{41}$$

where the Jacobian is:

$$A(t) = J\nabla^2 H(\boldsymbol{x}(t)) \tag{42}$$

with the structure[1]:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ -H_{xx} & -H_{xy} & 0 & 1 \\ -H_{yx} & H_{yy} & -1 & 0 \end{bmatrix} \tag{43}$$

where the Hessians of the Hamilton equation are:

$$H_{xx} = (1-\mu)\Big(\frac{1}{r_1^3} - 3\frac{(x+\mu)^2}{r_1^5}\Big) + \mu\Big(\frac{1}{r_2^3} - 3\frac{(x-1+\mu)^2}{r_2^5}\Big) \tag{44}$$

$$H_{yy} = (1-\mu)\left(\frac{1}{r_1^3} - 3\frac{y^2}{r_1^5}\right) + \mu\Big(\frac{1}{r_2^3} - 3\frac{y^2}{r_2^5}\Big) \tag{45}$$

$$H_{xy} = -3(1-\mu)\frac{(x+\mu)y}{r_1^5} - 3\mu\frac{(x-1+\mu)y}{r_2^5} \tag{46}$$

Using this method, it can be proven that A is a symplectic matrix with:

$$A = J\nabla^2 H, \quad (\nabla^2 H)^T = \nabla^2 H \tag{47}$$

$$\Phi(t)^T J\Phi(t) = J \tag{48}$$

After substituting $H_{xx} = 1 - U_{xx}$, $H_{yy} = 1 - U_{yy}$ and $H_{xy} = -U_{xy}$, the Lagrangian form of the STM is found to be:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ U_{xx} & U_{xy} & 0 & 2 \\ U_{yx} & U_{yy} & -2 & 0 \end{bmatrix} \tag{49}$$

This Lagrangian form is used to find the STM for the capture orbits in this paper.

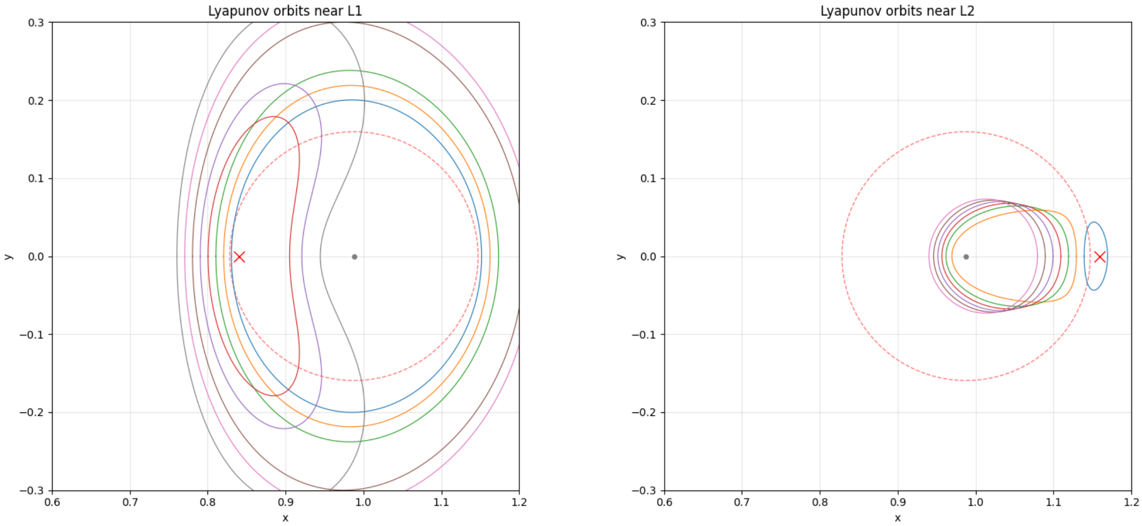

Figure 3: Lyapunov Orbits near L_1 and L_2

4.4.2 The L_1 and L_2 Equilibrium Points

Based on research by Paez and Guzzo, [4], slow close encounters of a third body captured by the secondary mass can be studied using the dynamics at the Lagrangian points L_1, L_2 of the CR3BP. This system is evaluated in the neighborhoods of these equilibrium points while the eigenvalues of the STM are evaluated to understand the stability of the local system. The Hamiltonian structure guarantees that the eigenvalues come in $(\lambda, -\lambda, \overline{\lambda}, -\overline{\lambda})$ pairs[3]. At L_1 and L_2 there are one real pair, $\pm\alpha$, and one imaginary pair $\pm i\beta$ each corresponding to hyperbolic (unstable) and center (stable) directions, respectively[6].

4.4.3 Lyapunov Orbits

The Lyapunov orbits are planar, periodic orbits that reside inside L_1 and L_2 and exist for Jacobi constants just below the neck-opening threshold[1]. They play a central role in studying temporary lunar capture because their invariant manifolds define the pathways connecting Earth-centered to Moon-centered motion. For Jacobi constants near the neck-opening values access to the lunar region is possible only through the L_1 and L_2 gateways. The stable and unstable manifolds govern the transport through these regions.[5] Temporary capture happens when a trajectory evolves in the stable manifold for a finite time before escaping on the corresponding unstable manifold[1]. Figure 3 shows the Lyapunov orbits near L_1 and L_2 with the dotted red circle representing the Moon's Hill sphere. The unstable manifolds are found by integrating forward in time while the stable manifolds are found by integrating backward in time. The manifolds of the largest Lyapunov orbit is shown in figure 4.

5 Simulation and Results

5.1 Determination of Initial Conditions

The ZVCs of various values of Jacobi constants were examined to understand potential initial conditions for the simulation of temporary moon capture, similar to the approach to Melo, et al.[2]. In figure 5, the three constants were chosen at the points where the L_1 neck begins to open (3.1885),

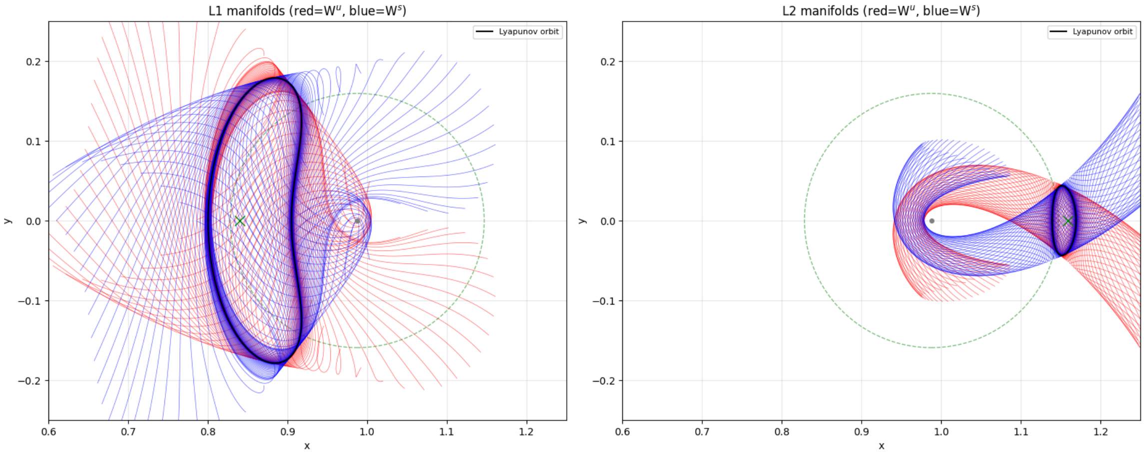

Figure 4: Lyapunov Manifolds near L_1 and L_2red=W^s, blue=W^u

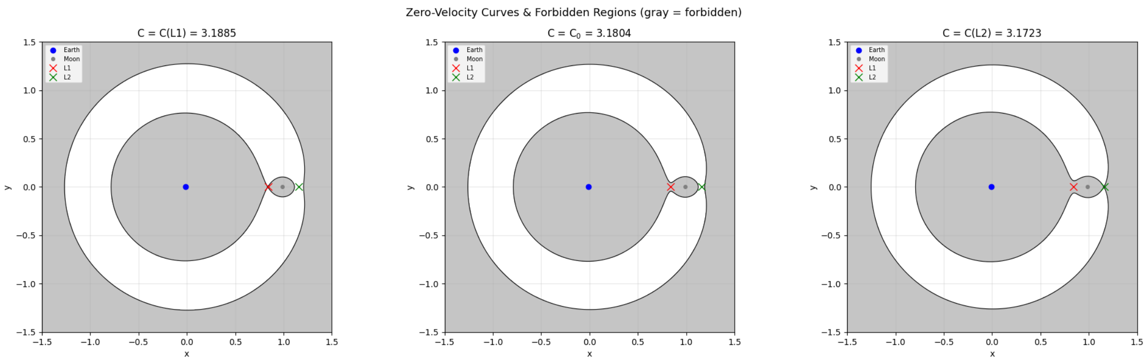

Figure 5: Forbidden Regions for Three Jacobi Constants

the L_2 neck begins to open (3.1723) and an intermediate value (3.1804). As the Jacobi constant decreases, the forbidden region becomes smaller, allowing more space between the Earth and Moon to be accessible to the third body. It is clear that an object cannot enter the Moon's Hill sphere without passing through the L_1 or L_2 neck. When the L_2 neck is open, escape from the Earth-Moon system is possible. The sweet spot for temporary capture without escape is when L_1 is open and L_2 is closed.

To explore orbit behavior in each of these regions, the initial position of the third body was slightly perturbed off the L_1 equilibrium point with velocity derived from equation 30 using the equation for the effective potential, equation 29. The simulated orbits are overlaid on the zero-velocity curves and forbidden regions in figure 6. The figure illustrates the behavior in the three regimes:

- **Left** - $C_j = C_J(L_1)$: The L_1 neck is barely open so the trajectory is confined to the Earth-side region and cannot reach the Moon.
- **Middle** - C_J=3.1804: The L_1 neck is open so the 3rd body passes through and gets temporarily captured, looping around the Moon inside the Hill sphere.
- **Right** - $C_J = C_J(L_2)$: Both L_1 and L_2 necks are open so the trajectory has even more freedom, entering and exiting the Moon region through both gateways with chaotic behavior.

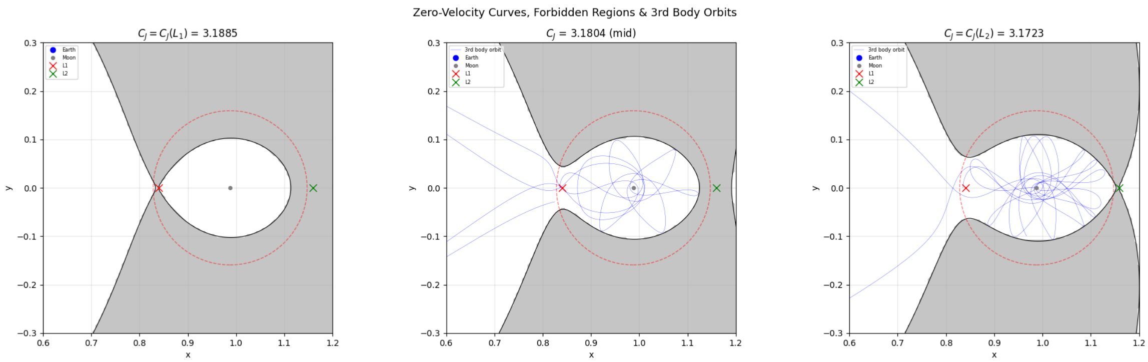

Figure 6: Simulated Trajectories for Third Body with Three Jacobi Constants
Panels correspond to C=C(L_1), C=3.1804, C=C(L_2)

5.2 Calculations and Verification of Numerical Results

Orbits transferring to the Moon's Hill sphere through the L_1 stable manifolds were investigated by reverse integrating along the associated unstable manifold. This allows the projection of the trajectory from where it came in the Moon's Hill sphere. The state transition matrix was calculated with Φ_0=I_{4x4} as the initial condition using Python's DOP853 library which uses an explicit Runge-Kutta method of order 8. The Lagrangian form of the dynamics matrix, A, in equation 49, was used which corresponds to a rotating frame. Because the state transition matrix is integrated in Lagrangian variables $(\boldsymbol{q}, \dot{\boldsymbol{q}})$, it is not symplectic in general. To verify the Hamiltonian structure of the solution flow, the STM was mapped to the canonical variables $(\boldsymbol{q}, \boldsymbol{p})$ using the Legendre transformation $\boldsymbol{p} = \frac{\partial L}{\partial \dot{\boldsymbol{q}}}$ and $\boldsymbol{q} = \boldsymbol{q}$. This resulted in a similarity transformation of:

$$T(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \begin{bmatrix} I_{4x4} & 0 \\ \frac{\partial^2 L}{\partial \dot{q} \partial q} & \frac{\partial^2 L}{\partial \dot{q}^2} \end{bmatrix} \tag{50}$$

For the planar CR3BP, $\frac{\partial^2 L}{\partial \dot{q}^2}$=$I_{2x2}$ and

$$\frac{\partial^2 L}{\partial \dot{q} \partial q} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{51}$$

which takes into account the Coriolis coupling. This results in the final transformation matrix:

$$T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \tag{52}$$

This transformation $\Phi_H = T \Phi_L T^{-1}$ was applied to the Lagrangian STM to recover the Hamiltonian form which was then used to assess symplecticity, confirming that $\Phi^T J \Phi = J$ where J is the canonical symplectic matrix defined in equation 36[3]. In addition, the determinant of Φ was verified to be unity.

5.3 Presentation of Results

Hundreds of samples were collected across multiple Lyapunov orbits (3,4 and 7) to determine if they exhibited capture behavior. Figure 7 shows a sampling of the trajectories that were successfully

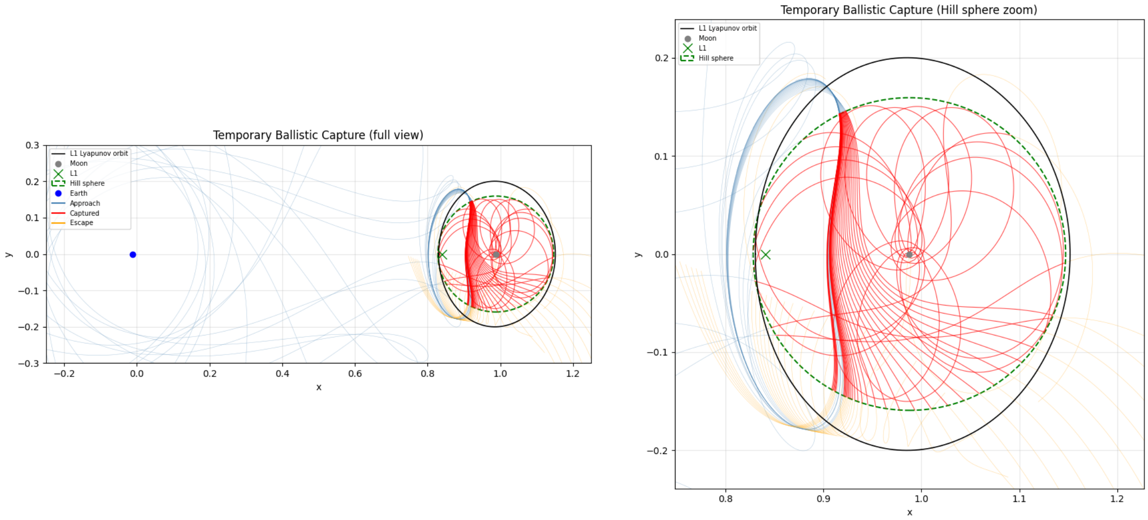

Figure 7: Sampling of Trajectories Originating Near L_1 red=captured, blue=entry,orange=escape

Orbit Family	**Sample Count**	**Mean Capture Duration Δt**	**Mean STM Growth Factor G_Φ**	**Mean Exp Growth Rate κ**
3	73	1.024	51.29	1.802
4	77	0.757	5.038	1.632
7	80	0.672	4.728	2.044

Table 1: Summary of Non-dimensional Data of Captured Trajectories

captured. The light blue lines show the trajectories prior to capture, the red lines show the trajectory inside the lunar sphere of influence and the orange lines show the trajectories after escape. The third body can escape through the L_1 or L_2 neck depending on its energy.

Overall there were two hundred thirty successful captures, seventy-three that originated in the third Lyapunov orbit, seventy-seven in the fourth and eighty in the seventh. Table 1 summarizes the results for each orbit family including the number of samples used, the mean duration of the captured orbit, the mean STM growth factor, we defined as:

$$G_\Phi = \frac{||\Phi||_{exit}}{||\Phi||_{entry}} \tag{53}$$

and the mean exponential growth rate:

$$\kappa = \frac{1}{\Delta t} ln(G_\Phi) \tag{54}$$

6 Discussion and Future Investigation

Temporary lunar capture occurs when trajectories evolve near the center manifold of the L_1 Lyapunov orbits for finite time before divergence along the unstable manifold leads to escape. The correlation between growth factor and capture time is 0.658 for the sampled trajectories considered here. This indicates that longer capture intervals are associated with larger STM growth factor, G_Φ, over the capture period. This indicates that extended residence time in the Moon's Hill sphere often coincides with substantial accumulated divergence away from the stable manifold. The unstable STM eigenvalue sets the exponential rate of divergence from the capture region, directly linking local linear instability to the observed duration of temporary capture. The next step in this investigation would be to deal with data outliers. There are a few samples where the magnitude of the STM at exit is on the order of 1.06×10^7. These are either highly unstable passageways or accumulated numerical errors. If this analysis is sensitive to large tails, the results could be skewed. After that, the effects of the Sun should be included in the dynamics to understand its impact on capture duration.

7 Conclusion

Determining the set of trajectories that result in temporary capture of a third body by the Moon is greatly simplified by exploiting the dynamical structure near the L_1 Lagrange point. By using perturbations about Lyapunov orbits as initial conditions, the search for capture trajectories can be restricted to a well-defined region of phase space organized by invariant manifolds. In particular, the center manifold associated with the saddle–center equilibrium provides a natural framework for identifying trajectories that exhibit prolonged residence within the lunar region. Integrating these perturbed states both forward and backward in time reveals families of trajectories that enter the Moon's vicinity along stable manifolds and eventually escape along unstable manifolds, consistent with the Hamiltonian geometry of the CR3BP. Analysis of the State Transition Matrix further shows that temporary capture is governed by inherent instability near L_1, with capture duration influenced by the rate of divergence of nearby trajectories. Together, these results demonstrate that temporary lunar capture can be understood as a manifestation of phase-space transport near Lyapunov orbits, providing a systematic and physically meaningful approach for identifying capture and escape pathways in the Earth–Moon system.

8 References

[1] W.S Koon and M.W. Lo. *Dynamical Systems, the Three-Body Problem and Space Mission Design*. Mission Design Book Online, 2022.

[2] C.F. de Melo and E.E.N Macau. "Numerical study about natural escape and capture routes by the Moon via Lagrangian points L1 and L2". In: *Advances in Space Research, Vol. 40, pp 83-95* (2007).

[3] K.R. Meyer and D.C. Offin. *Introduction to Hamiltonian Dynamical Systems and the N-Body Problem*. Springer, 2017.

[4] R.I. Paez and M. Guzzo. "Astudy of temporary captures and collisions in the Circular Restricted Three-Body Problem with normalizations of the Levi-Civita Hamiltonian". In: *International Journal of Non-Linear Mechanics* (2020).

[5] R. P. Russell and Try Lam. "Designing Ephemeris Capture Trajectories at Europa Using Unstable Periodic Orbits". In: *Journal of Guidance, Control and Dynamics, Vol. 30, No. 2* (2007).

[6] V. Szebehely. *Theory of Orbits: The Restricted Problem of Three Bodies*. Academic Press, 1967.